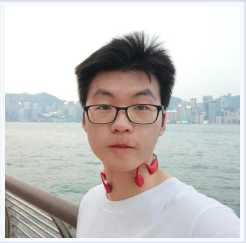
RESEARCH CV

Zhuohang Jiang

LLMs · AI Agents · Multimodal RAG · Reasoning

(+86) 152-6704-1093 | zhuohang.jiang@connect.polyu.hk
zhuohangjiang.com | GitHub: jzzzzh | Google Scholar



Research Profile

LLM reasoning, multimodal RAG, self-evolving agents. **8 published/accepted contributions; 7 at CCF-A venues/related tracks, 5 first/co-first authored**: KDD, CVPR, NeurIPS, AAAI. **633 citations; h-index: 7**.

Education

The Hong Kong Polytechnic University | Ph.D. in Computer Science Sep. 2024 – Present
Advisors: Prof. Qing Li and Prof. Wenqi Fan; GPA: **3.60/4.00**.
Sichuan University | B.Eng. in Computer Science and Technology Sep. 2020 – Jun. 2024
Major GPA: **3.79/4.00**; overall GPA: 3.78/4.00.

Industry Experience

Kuaishou Technology | Research Intern, Short-Video E-commerce Oct. 2025 – Present

- Developed **AB Agent**: structured strategy/code changes, parameters, and outcomes into retrievable units; integrated A/B comparisons, tuning recommendations, configuration checks, and review. Achieved **GMV +4.829%** with positive guardrails and continuous experience updates.
- Developed **AIR**: decomposed LLM-derived intent semantics into reusable atomic knowledge; approximated sequence-level reasoning through online retrieval/composition, reducing real-time LLM costs.

Selected Research

[KDD'25] HiBench | Hierarchical Structure Reasoning in LLMs
First Author / Project Lead · KDD 2025, Benchmark & Dataset Track
Problem: Existing benchmarks neglect hierarchical dependencies. **Method:** Led benchmark/code design; built controllable tree/JSON/formula generators, algorithmic labels, and code/paper tasks (**30 tasks, 39,519 queries**); compared representations/complexity/CoT across **20 LLMs / 10 families**. **Results:** Exposed implicit-hierarchy weaknesses; instruction tuning yielded relative gains of **88.84% (Llama-3.1-8B)** and **31.38% (Qwen2.5-7B)**; released data/toolkit.

[CVPR'26] SUPERGLASSES / SUPERLENS | Smart-Glasses VLM Benchmark & Agent
First Author · CVPR 2026 Findings
Problem: Smart-glasses QA requires small-object grounding and external knowledge absent from standard VQA. **Method:** Built **2,422 real-device QA pairs**; routed direct answers/RAG, localized objects, decomposed queries, cleaned HTML, and reranked image/text evidence. **Two-layer Redis caching** reuses search/HTML results. **Results:** Across **26 VLM/RAG variants**, SUPERLENS improved Qwen2.5-VL-7B from **32.82 to 44.10 (+11.28 points)**.

A/B Agent | Self-Evolving Agent for Industrial Strategy Iteration
First Author · arXiv:2608.04625, 2026 (Preprint)
Problem: Flat RAG cannot reuse fragmented experiment knowledge for iterative tuning. **Method:** Built experience trees with **sparse/dense retrieval**, tree-path boosts, and **Qwen-Reranker** for evidence-backed configurations. Compared **parent–child/sibling experiments**, confidence-weighted gains, and guardrail penalties for local tuning/branch switching and memory updates. **Results:** Achieved **GMV +4.829%**, with positive gains across all guardrails.

[KDD'26] AIR | Composable LLM Intent Reasoning with Offline-to-Online Retrieval
First Author · KDD 2026, Applied Data Science Track
Problem: Online LLM costs prevent real-time reasoning over long, noisy histories. **Method:** Cached LLM-derived atomic intent paths from behavior/user/object context; composed intent trees with action weights/time decay, retrieved target-relevant chains, and fused evidence via **multi-head attention**, without online LLM calls. **Results:** Approximately **400×** **throughput** vs. online LLM calls; **GMV +3.446%**.

[KDD'25] QA-Dragon | Query-Aware Dynamic RAG for Knowledge-Intensive VQA
First Author · KDD Cup 2025 Workshop on Multimodal Retrieval Augmented Generation
Problem: Fixed retrieval mismatches VQA evidence needs. **Method:** Routed drafts to **direct output / search verification / RAG**; selected image/text tools, cropped targets, reranked and verified evidence. **Results:** Single-/multi-source/multi-turn gains: **5.06% / 6.35% / 5.03%**.

Academic Activities, Honors & Skills

Activities: KDD 2025 Reasoning Day invited talk; KDD Cup 2025 CRAG-MM, 12th globally (team lead); NeurIPS 2024 Spotlight (co-first author).
Skills: PyTorch, Hugging Face, RAG, agent systems, Docker, Git; Mandarin (native), English (IELTS 6.5).